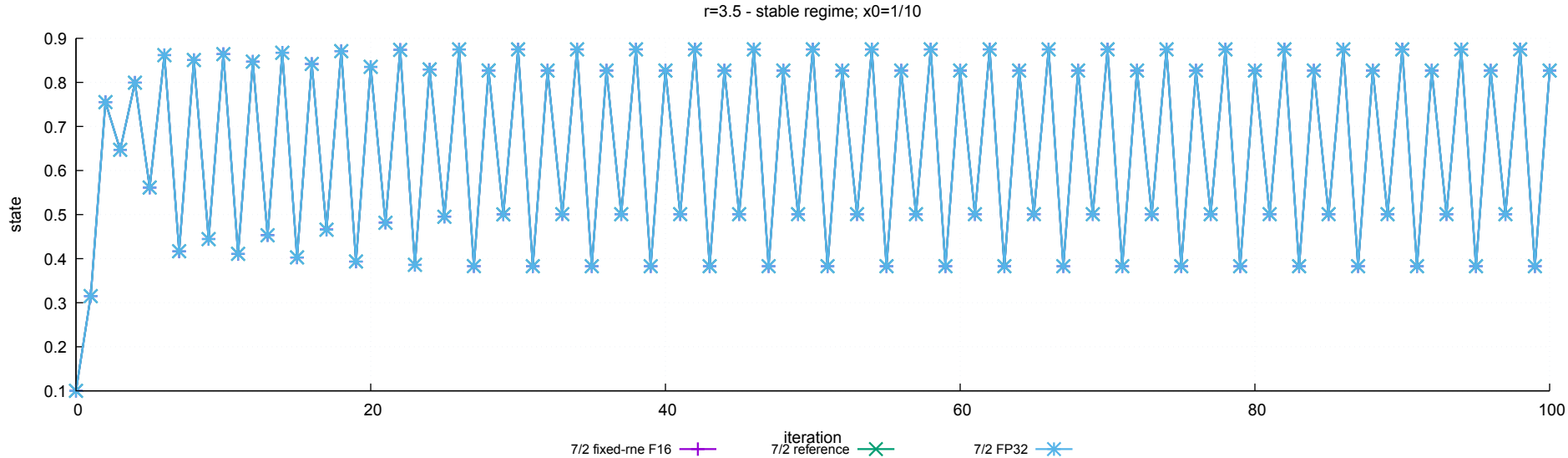
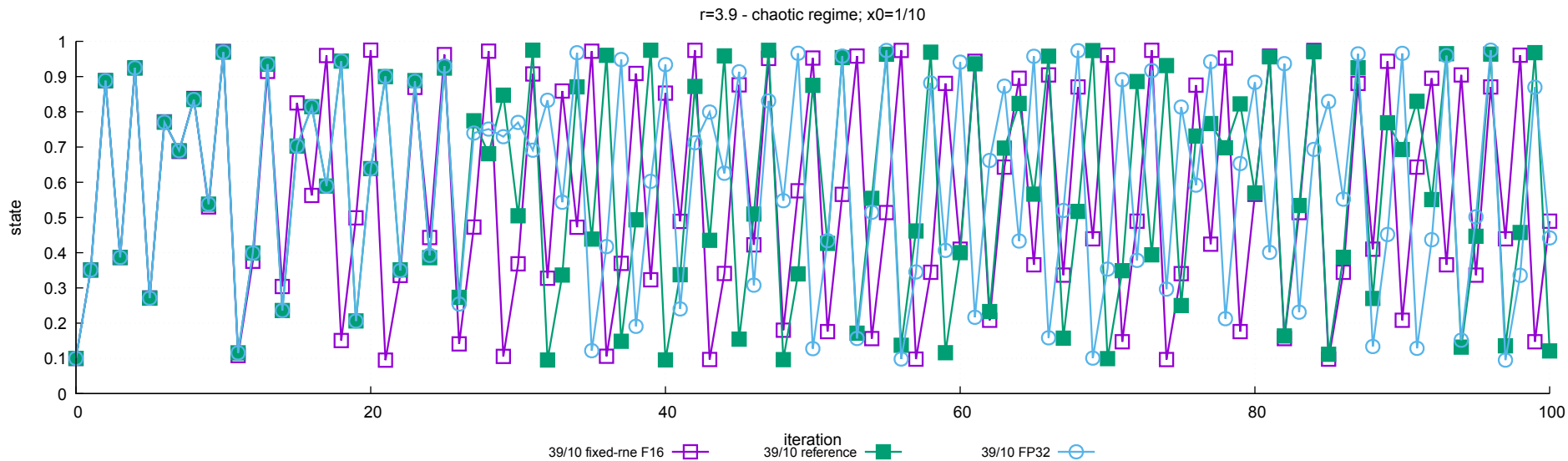
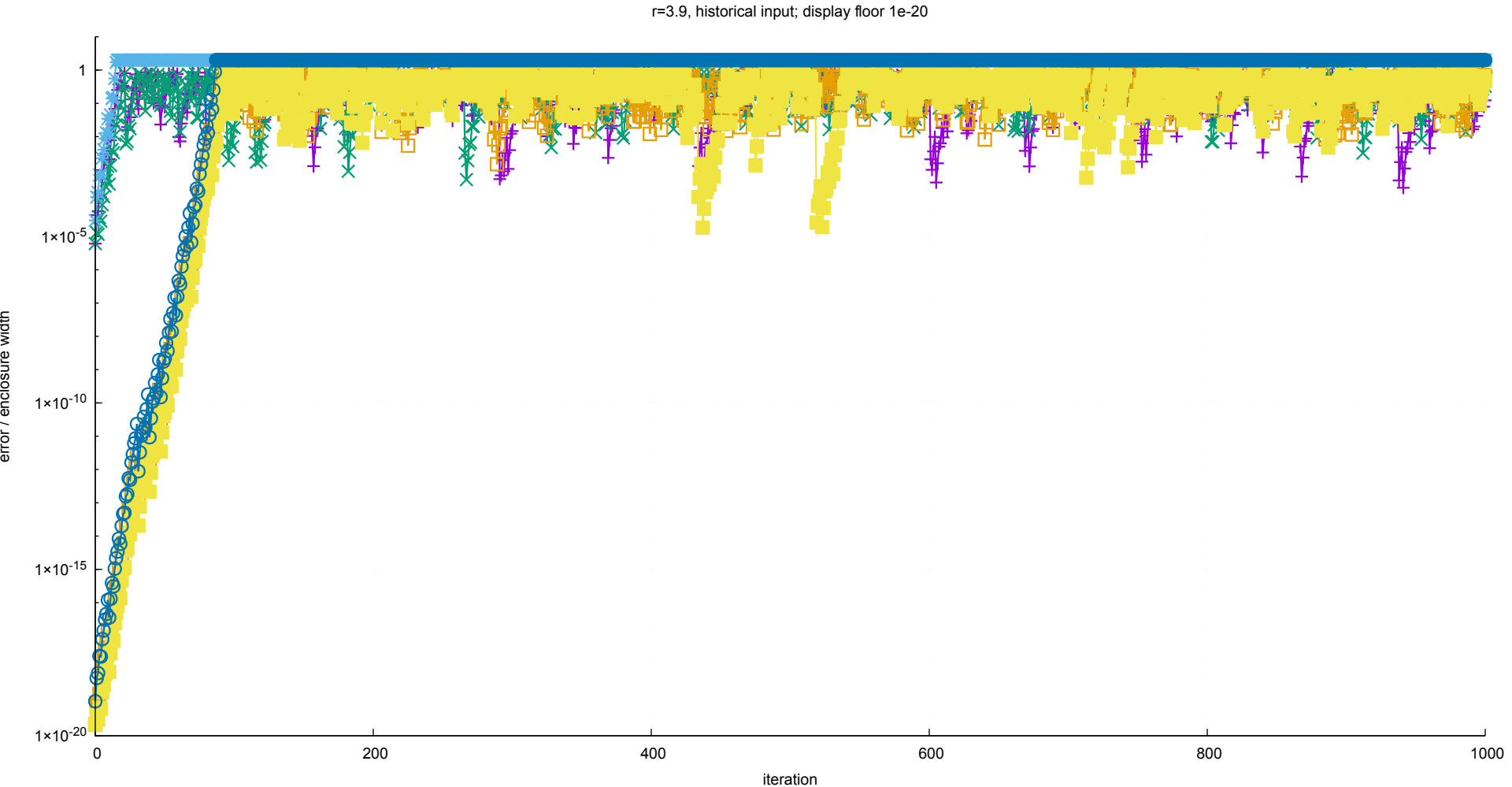
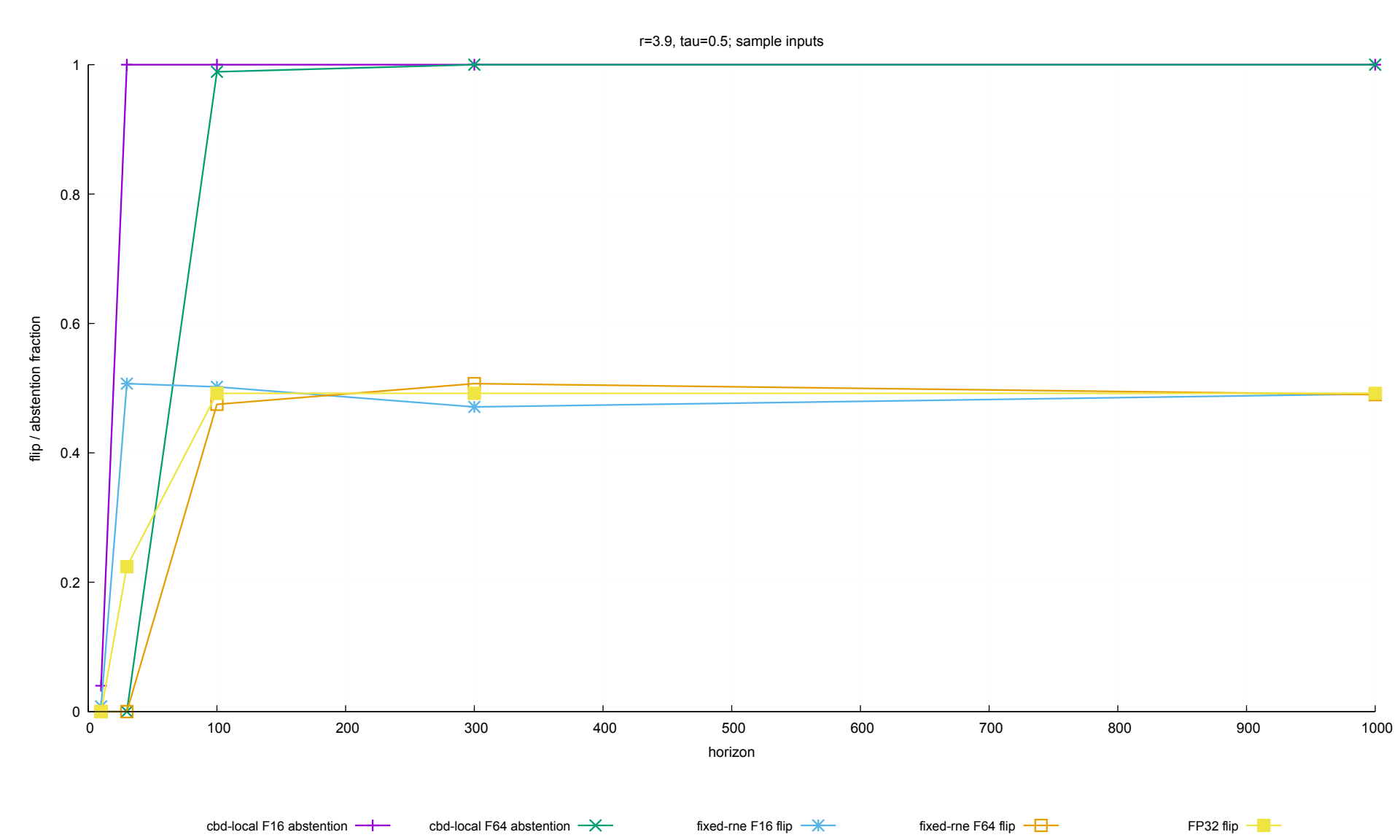
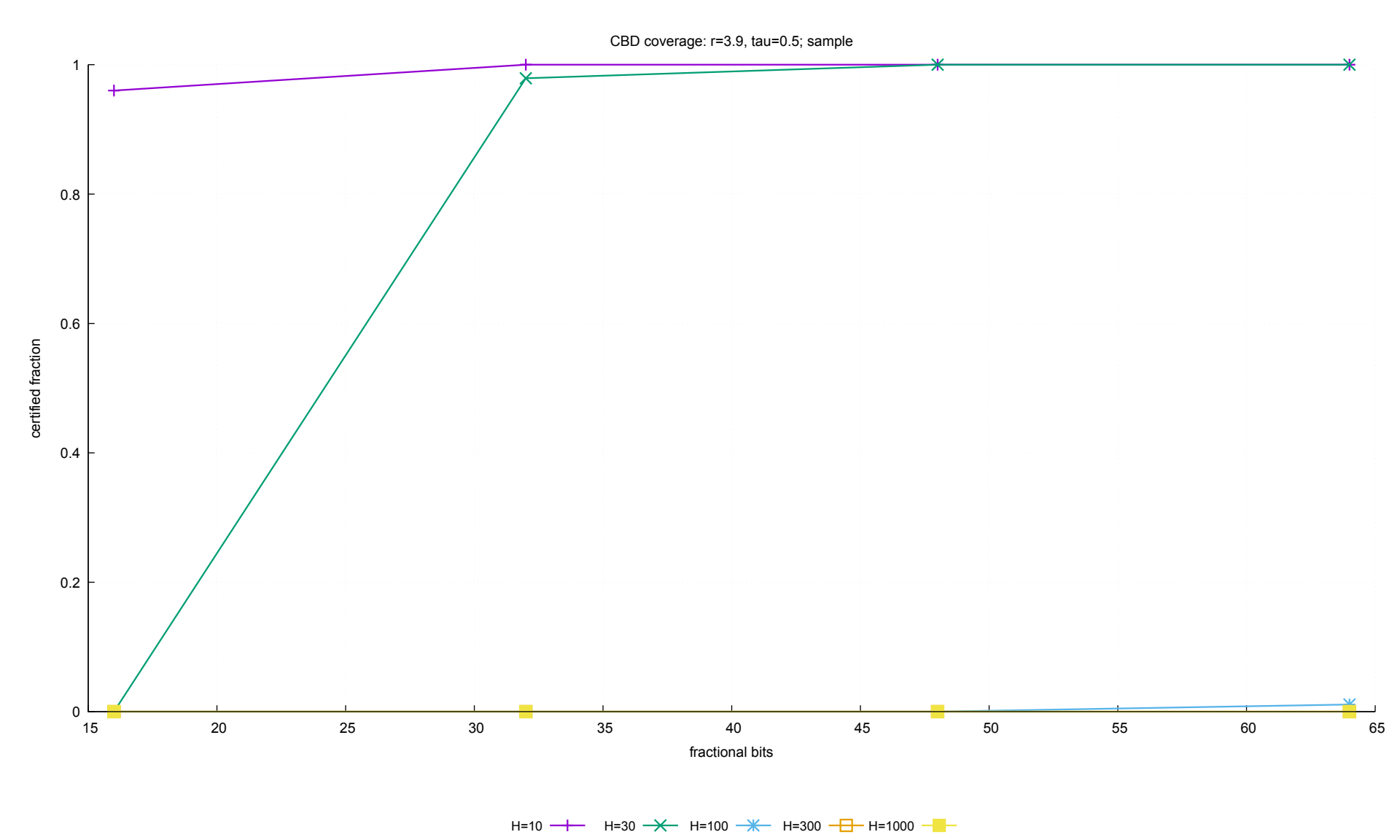
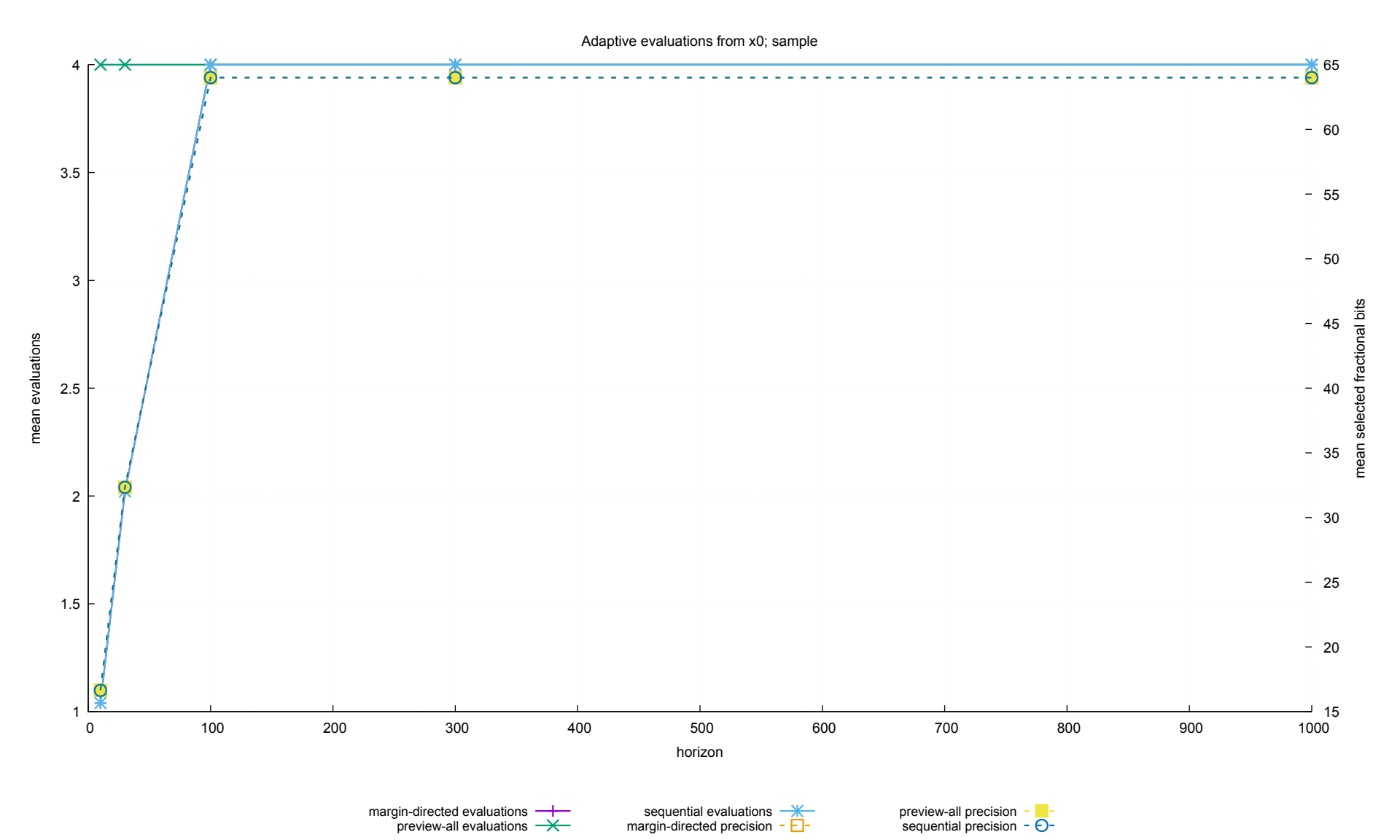

r=3.5 - stable regime; x0=1/10
state
iteration
7/2 fixed-rne F16
7/2 reference
7/2 FP32

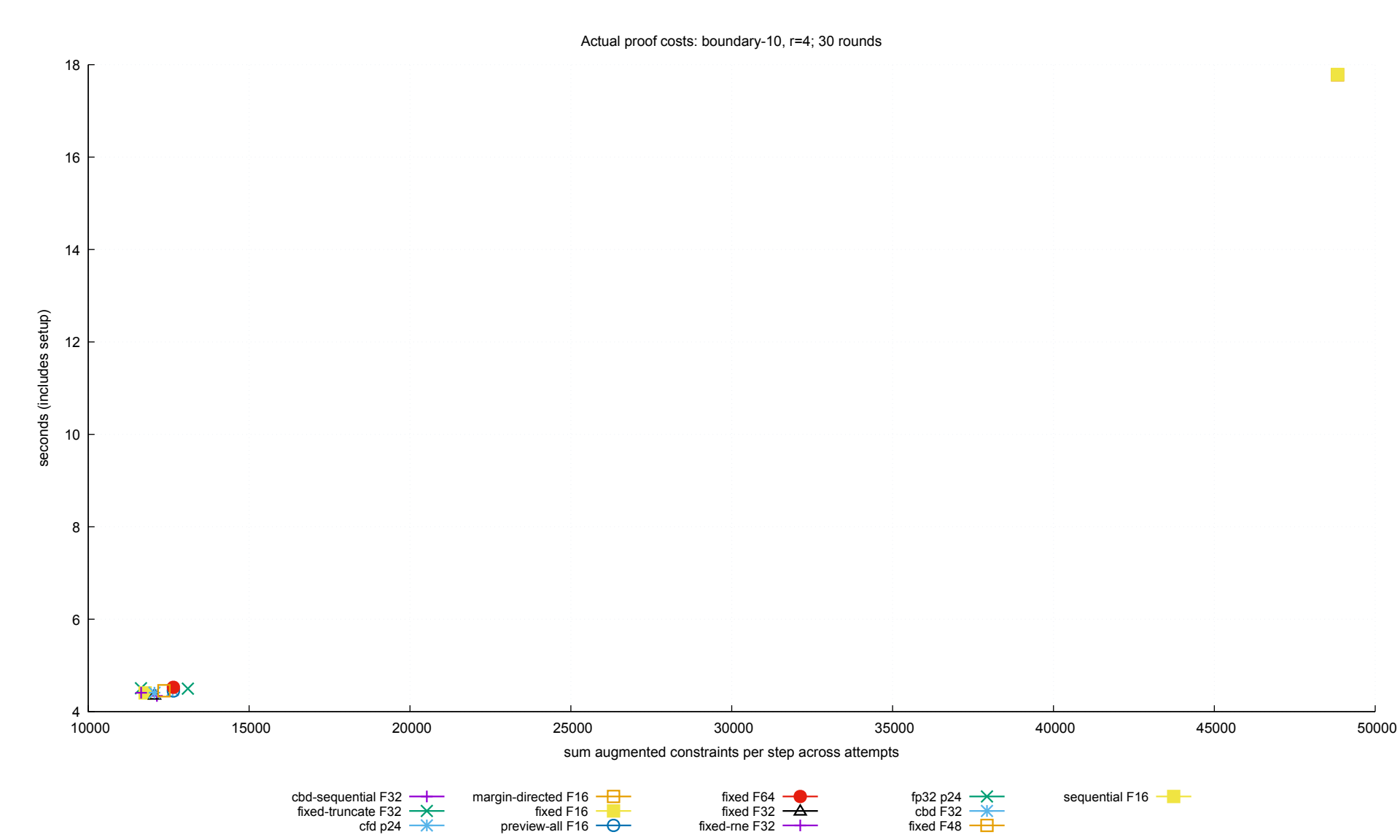

r=3.9 - chaotic regime; x0=1/10
state
iteration
39/10 fixed-rne F16
39/10 reference
39/10 FP32

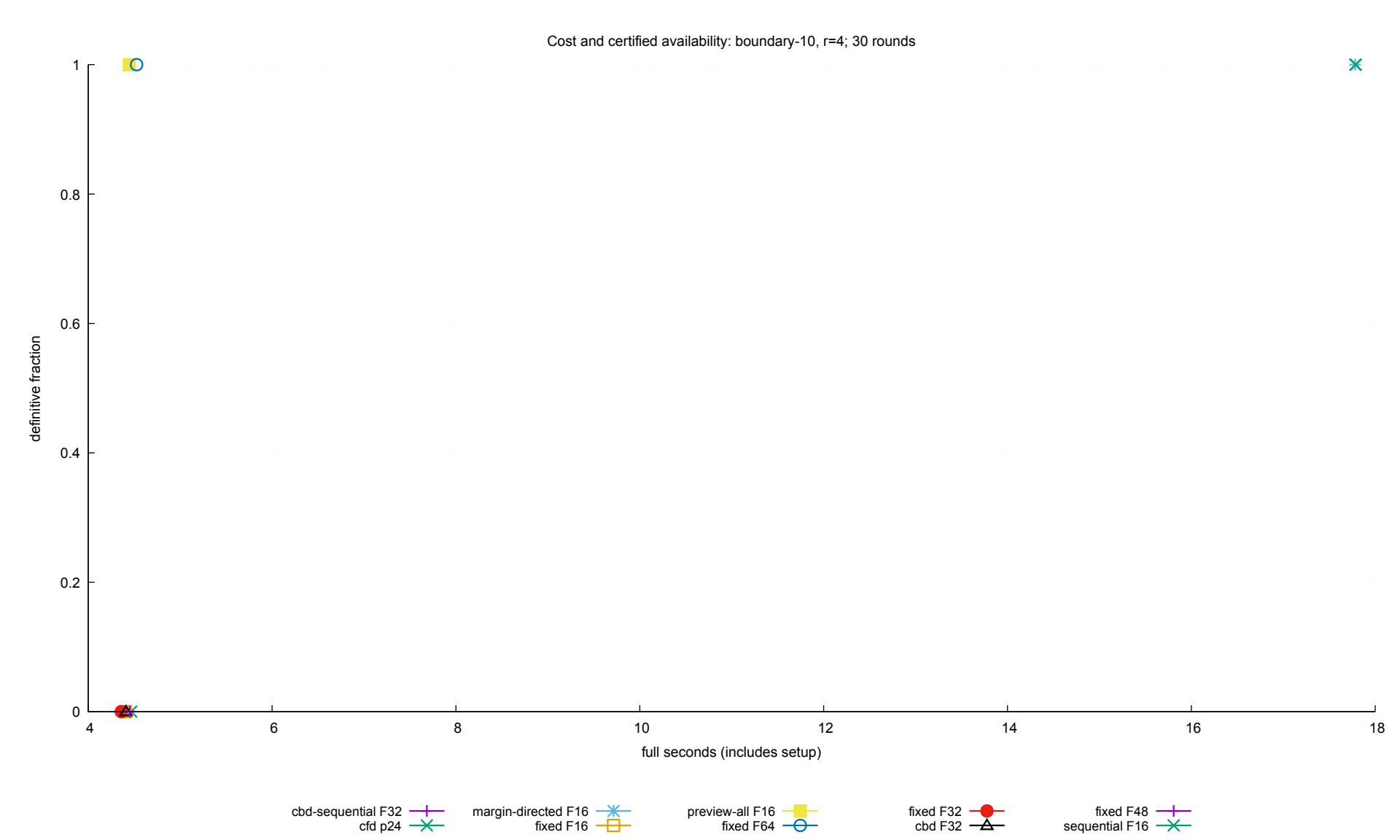

r=3.9, historical input; display floor 1e-20
error / enclosure width
1
1×10-5
1×10-10
1×10-15
1×10-20
0
200
400
600
800
1000
iteration
fixed-rne F16 center error
cbd-local F16 center error
CBD F16 width
fixed-rne F64 center error
cbd-local F64 center error
CBD F64 width


r=3.9, tau=0.5; sample inputs
flip / abstention fraction
horizon
0
0.2
0.4
0.6
0.8
1
0
100
200
300
400
500
600
700
800
900
1000
cbd-local F16 abstention
cbd-local F64 abstention
fixed-rne F16 flip
fixed-rne F64 flip
FP32 flip


CBD coverage: r=3.9, tau=0.5; sample
certified fraction
fractional bits
0
0.2
0.4
0.6
0.8
1
15
20
25
30
35
40
45
50
55
60
65
H=10
H=30
H=100
H=300
H=1000


Adaptive evaluations from x0; sample
mean evaluations
mean selected fractional bits
horizon
4
3.5
3
2.5
2
1.5
1
65
60
55
50
45
40
35
30
25
20
15
0
100
200
300
400
500
600
700
800
900
1000
margin-directed evaluations
preview-all evaluations
sequential evaluations
margin-directed precision
preview-all precision
sequential precision


Actual proof costs: boundary-10, r=4; 30 rounds
seconds (includes setup)
sum augmented constraints per step across attempts
18
16
14
12
10
8
6
4
10000
15000
20000
25000
30000
35000
40000
45000
50000
cbd-sequential F32
fixed-truncate F32
cfd p24
margin-directed F16
fixed F16
preview-all F16
fixed F64
fixed F32
fixed-rne F32
fp32 p24
cbd F32
fixed F48
sequential F16


Cost and certified availability: boundary-10, r=4; 30 rounds
definitive fraction
full seconds (includes setup)
0
0.2
0.4
0.6
0.8
1
4
6
8
10
12
14
16
18
cbd-sequential F32
cfd p24
margin-directed F16
fixed F16
preview-all F16
fixed F64
fixed F32
cbd F32
fixed F48
sequential F16